Random Variables

A random variable X assumes any of several different numeric values as a result of a random event. If X takes on one of countably many distinct outcomes, then X is discrete.

A probability model associates a probability P to each value of X: $P(X=x)$.

Ex. Consider a certain policy: the company

- pays \$10000 if you die,
- pays \$5000 if you are disabled,
- charges \$50 a year.

Suppose that, every year, $\frac{1}{1000}$ will die, and $\frac{2}{1000}$ will be permanently disabled.

Q. Is \$50 a good price?

Let X be the amount the company pays out each year.

Outcomes	Payouts x	$P(X=x)$
Death	10000	$1/1000$
Disabled	5000	$2/1000$
Neither	0	$997/1000$

For each policy each year,

$$10000 \cdot \frac{1}{1000} + 5000 \cdot \frac{2}{1000} + 0 \cdot \frac{997}{1000}$$

$$= 10 + 10 + 0 = \$20.$$

This is the expected value of X:

$$\mathbb{E}[X] = \sum x \cdot P(X=x).$$

Outcomes	~~Payouts x~~ Deviations $x - \mathbb{E}[X]$	$P(X=x)$
Death	$10000 - 20 = 9980$	$1/1000$
Disabled	$5000 - 20 = 4980$	$2/1000$
Neither	$0 - 20 = -20$	$997/1000$

We find the average deviation:

$$\sqrt{9980^2 \cdot \frac{1}{1000} + 4980^2 \cdot \frac{2}{1000} + (-20)^2 \cdot \frac{997}{1000}} \approx 386\ldots$$

This is the standard deviation:

$$\sigma = \sqrt{\sum (x - \mathbb{E}[X])^2 \cdot P(X=x)}.$$

variance

Proposition 1. For any $c \in \mathbb{R}$:

$$\mathbb{E}[X \pm c] = \mathbb{E}[X] \pm c,$$

$$\mathrm{Var}(X \pm c) = \mathrm{Var}(X).$$

Proposition 2. For any $a \in \mathbb{R}$:

$$\mathbb{E}[aX] = a\,\mathbb{E}[X],$$

$$\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X).$$

Proposition 3. Given discrete random vars.,

$$\mathbb{E}[X \pm Y] = \mathbb{E}[X] \pm \mathbb{E}[Y]$$

If X, Y are independent:

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Food for Thought. Is $X_1 + X_2 = 2X$?

Uniform Distribution

All events are equally likely to occur!

To describe a uniform distribution, it is enough to know the number of outcomes.

Let X be a uniformly distributed discrete random variable:

$$X \sim U(n).$$

parameter: # of outcomes

For all x_i,

$$P(X = x_i) = \frac{1}{n}.$$

We can compute:

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \cdot \underbrace{P(X = x_i)}_{\text{always } \frac{1}{n}} = \frac{1}{n} \cdot \sum_{i=1}^{n} x_i.$$

$$\mathrm{Var}(X) = \frac{1}{n} \cdot \sum_{i=1}^{n} (x_i - \mathbb{E}[X])^2.$$

Bernoulli Trial

Opening a blind box is a trial:

- two outcomes; secret vs. not secret
- fixed probability of success; $p = 1/72$
- trials are independent.

Let X be a Bernoulli random variable:

x	$P(X=x)$
0	$1-p$
1	p ← parameter: probability of success

We compute:

$$\mathbb{E}[X] = 0\cdot(1-p) + 1\cdot p = p$$

$$\begin{aligned}\mathrm{Var}(X) &= (0-p)^2\cdot(1-p) + (1-p)^2\cdot p \\ &= p^2(1-p) + (1-p)^2 p \\ &= p(1-p)(p+1-p) = p(1-p).\end{aligned}$$

Ex. Do the following experiments involve Bernoulli trials?

- Roll 50 dice, list out all rolls. NO!
- In a survey, we find that 10% of first-years study Mathematics Yes...
- Draw seven hearts from a deck of cards. NO!
- Apple realizes that 1% of the iPhone Airs they released are missing a camera. Yes...

Geometric Distribution

Consider a sequence of Bernoulli trials. We model how long it takes to achieve the first success.

parameter

If p is the probability of succeeding:

$$P(X = x) = (1-p)^{x-1} p.$$

Here, X is the discrete random variable that counts the number of trials required until the first success.

The expected value and variance are:

$$\mathbb{E}[X] = \frac{1}{p}, \quad Var(X) = \frac{1-p}{p^2}.$$

Ex. The probability of getting the "secret" is $1/72$.

- What is the probability that we find it in the second box opened?

$$\left(\frac{71}{72}\right) \cdot \left(\frac{1}{72}\right) \approx 0.0137$$

- Each box costs \$24.90. What is the expected value of a person's spending to obtain the toy? 24.90×72

Let Y be the amount spent. Then, $Y = 24.9X$, where X is the number of boxes opened. So:

$$\mathbb{E}[Y] = \mathbb{E}[24.9X] = 24.9\,\mathbb{E}[X]$$

Since $\mathbb{E}[X] = \frac{1}{1/72} = 72$,

$$\mathbb{E}[Y] = 24.9 \cdot 72 \approx 1794.8.$$

* $Var(Y) \approx 3169491.12\ldots \Rightarrow \sigma \approx 1780.$

Binomial Distribution

We model the probability of achieving a certain number of successes within a fixed number of trials.

We require two parameters:

- number of trials n,
- probability of success p.

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x},$$

where $\binom{n}{x} = \dfrac{n!}{x!\,(n-x)!}$. ← # of ways to select x out of n

Let's compute $\mathbb{E}[X]$: view X as a collection of n Bernoulli trials $X = X_1 + X_2 + \cdots + X_n$.

$$\Rightarrow \mathbb{E}[X] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n]$$

$$= p + p + \cdots + p = np.$$

$$\Rightarrow \mathrm{Var}(X) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)$$

$$= p(1-p) + \cdots + p(1-p) = n \cdot p(1-p)$$

Ex. A company finds that 90% of all e-mails are spam. Suppose you have 25 messages in your e-mail.

- How many real messages do you expect?
- What is the probability that you only find 1 or 2 real messages?

Let X be the number of real messages.

Then, $X \sim \mathrm{Binom}(25, 0.1)$.

- $\mathbb{E}[X] = 25 \cdot 0.1 = 2.5$

  $\mathrm{Var}(X) = 25 \cdot 0.1 \cdot 0.9 \Rightarrow \sigma = 1.5$

- Find $P(X = 1 \cup X = 2)$

$$= P(X=1) + P(X=2)$$

$$= \binom{25}{1}(0.1)(0.9)^{24} + \binom{25}{2}(0.1)^2(0.9)^{23}$$

$$\approx 0.465\ldots$$

Poisson Distribution

In the early 1990s, a leukemia cluster was discovered in MA.

- US Population : 280,000,000
- Leukemia Cases : 30,800
- Town Population: 35,000

We estimate: the probability of getting leukemia is about $p = 30800/280000000 \approx 0.00011$.

For the town, we expect $35000 \cdot 0.00011 \approx 3.85$ cases.

Q. What is the probability of x cases?

$$\binom{35000}{x} (0.00011)^x (1-0.00011)^{35000-x}$$

This is very difficult to compute: n is huge and p is tiny!

parameter: mean!

We say that $X \sim \text{Pois}(\lambda)$:

$$P(X=x) = e^{-\lambda} \frac{\lambda^x}{x!}.$$

$$\Rightarrow E[X] = \lambda, \quad \text{var}(X) = \lambda$$

Poisson's Idea (n huge, p tiny)

Let $X \sim \text{Binom}(n,p)$, $E[X] = np$.

Construct a new model with the same expected value $\lambda = np$.

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{n(n-1)(n-2)\cdots(n-x+1)}{x!} \frac{\lambda^x}{n^x} \left(1-\frac{\lambda}{n}\right)^n \left(1-\frac{\lambda}{n}\right)^{-x}$$

Take a limit as $n \to \infty$:

- $\frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdots \frac{n-x+1}{n} \xrightarrow[n\to\infty]{\text{as}} 1$
- $\left(1-\frac{\lambda}{n}\right)^n \xrightarrow[n\to\infty]{\text{as}} e^{-\lambda}$

When $n \to \infty$,

$$P(X=x) \approx e^{-\lambda} \frac{\lambda^x}{x!}.$$

Poisson Distribution

This approximates a binomial distribution when n is large and p is small:

$$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}.$$

This model is completely determined by the parameter λ: the expected value.

Ex. I need to eat lunch, but I don't want to miss any students looking for me. Suppose that, on average, 2 students find me every hour.

- What is the probability that nobody comes in the next 30 mins?
- What is the probability that somebody comes in the next 30 mins?

Let X = # of students that find me in a 30-min. interval. Then:

$$X \sim \text{Pois}(1).$$

average # of students every 30 minutes

- What is the probability that nobody comes in the next 30 mins?

This is just:

$$P(X=0) = e^{-1} \frac{1^0}{0!} = \frac{1}{e} \approx 0.36788\ldots$$

- What is the probability that somebody comes in the next 30 mins?

This is

$$P(X \neq 0) = 1 - P(X=0) = 1 - \frac{1}{e} \approx 0.63212\ldots$$